

Advances in Physics

ISSN: 0001-8732 (Print) 1460-6976 (Online) Journal homepage: https://www.tandfonline.com/loi/tadp20

Recent magnetic investigations at Cambridge University

J.C. Belshé

To cite this article: J.C. Belshé (1957) Recent magnetic investigations at Cambridge University, Advances in Physics, 6:22, 192-193, DOI: 10.1080/00018739700101561

To link to this article: https://doi.org/10.1080/00018739700101561

Published online: 02 Jun 2006.

Submit your article to this journal

Article views: 28

View related articles

Citing articles: 2 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tadp20

Recent Magnetic Investigations at Cambridge University

By J. C. Belshé
Department of Geodesy and Geophysics, Cambridge

Recent in geological time as well as in the time of investigation are our studies of archaeological materials. Working with the Museum of Classical Archaeology, our department has concentrated its attention on the baked clay walls of ancient kilns. These we can collect *in situ*. The data are yielding us a curve for the secular variation of the magnetic field in England since the beginning of the Christian era.

Allied with this work is an experiment conducted on a reconstructed kiln of the Romano-British type. This was fired in an attempt to duplicate the ancient pottery technique. We viewed it as an opportunity to test our theories on the fixing of the magnetic vector in ancient kilns. The kiln was fired last month (October 1956). We studied temperatures obtained inside the kiln and in its walls, the change in magnetic moment of the kiln and its contents during firing, and the anomaly in the geomagnetic field caused by the kiln after its use. Samples were removed from the kiln wall for study in the laboratory in the manner done for archaeological remains. These do document the magnetic field directions at the place of firing. A report of this will be published next spring.

Our investigations of the field change in the vicinity of the kiln were novel in that they made use of the proton precession technique. We hope to use such an instrument in the future for locating buried kilns.

Mr. R. Girdler has studied the palaeomagnetism of some Jurassic rocks from Dorset and Devon. Though still in progress, this work shows a mean direction further from the present magnetic field than that of Nairn's which was reported at this conference. For example, the Bridport Sands have a mean direction of N 15° E with a downward dip of 54° and the Northants Iron-stones, N 19° W with a downward dip of 56°.

Studies of the remanent magnetization in the Dartmoor Granite show a very scattered remanent magnetic direction. The intensity of magnetization is low (10^{-6} to 10^{-7} e.m.u./cm^3) and the magnetic properties of the rock are quite variable. This is a preliminary study to investigations in the metamorphosed Culm measures of Carboniferous age lying north and east of the granite.

Finally I will mention some thermal studies done on Old Red Sandstones samples. These are from the Brecon Beacons area and were studied in conjunction with Carboniferous material from the South Wales coalfield. The work was done at Imperial College with the kind consent of Professor P. M. S. Blackett on a magnetometer-furnace apparatus

constructed by J. H. Leng. These Red Sandstones when cooled from above their Curie temperature in a field of 0·6 oe had a T.R.M. of the order of half their N.R.M. This contrasts sharply with the behaviour of similar material studied by Creer which had a N.R.M. a fraction of its T.R.M. This question must be posed : to what extent is this reported behaviour typical of material which may be suspected to have acquired a chemical magnetization ?